Multiple Access

Lecture 27

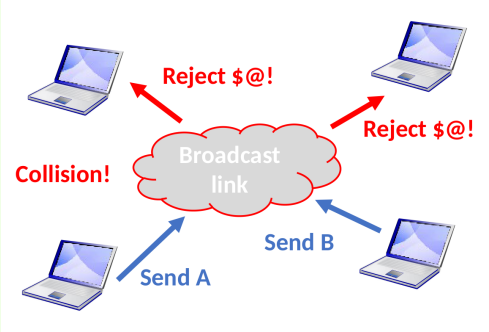
Computer Networks and Distributed Programming

Many machines, one wire

- Every link we have taken apart so far had two ends: one sender, one receiver, and no question about whose turn it was.
- A shared medium changes that. Whatever one station puts on it is heard by all the others, and two that transmit together destroy each other.
- There is no arbiter: no master, and no separate channel on which to ask permission. The rules must be agreed on the very channel being contended for.
- That is the multiple access problem, and the MAC sublayer is the part of the link layer that solves it.
- Three families of rules, and for each one what it buys and what it costs. Then the timing argument that shaped Ethernet.

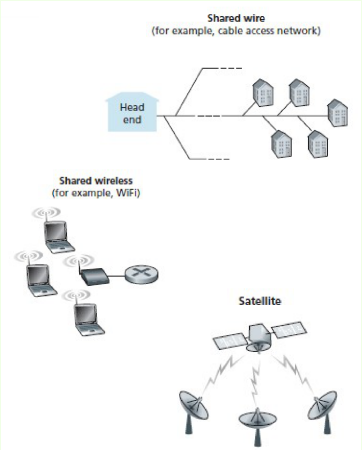
When two frames meet

- Two nodes transmit at the same time: the signals add up on the medium and no receiver can make sense of either frame.
- The garbled frames reach everybody and fail the check we built last time, so they are dropped as errors.
- First cost: every frame involved is lost, and every sender has to try again.
- Second cost: the channel was busy the whole time and carried nothing. That interval is gone.



Where shared channels are

- Shared wire: the coaxial cable of a cable access network, with the head end at one end and the houses along it.
- Shared wireless: every station in range of an access point transmits into the same air.
- Satellite: ground stations thousands of kilometres apart, all heard by the same transponder.
- Different technologies, one problem. Hundreds or thousands of nodes may sit on a single broadcast channel.



What we would want from the rules

Desiderata

A multiple access protocol for a broadcast channel of rate R should have four properties. No protocol has all four, and which one it gives up is what distinguishes one family from another.

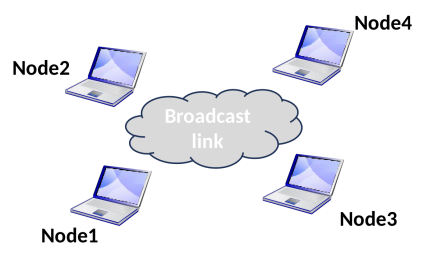
- When a single node has data to send, that node gets the full R: the channel is not slowed down by absent stations.
- When M nodes have data to send, each gets about R/M on average — not instant by instant, but over a reasonable interval.
- Decentralised: no master node, because a master is a single point of failure for the whole channel.
- Simple and cheap: the rules run on every frame, in hardware, so any overhead is paid millions of times.

Three ways to share a channel

- Channel partitioning: the capacity is cut up in advance and each node is given a piece of it, once and for all.
- Random access: nobody is given anything: a node transmits at the full rate, collides, and tries again.
- Taking turns: the right to transmit is passed around, so that only one node is ever entitled to the channel.
- They differ in when the arbitration happens: before the transmission, after it, or around it.
- Every real protocol is one of the three, or — as we will see at the end — all three in different parts of the same network.

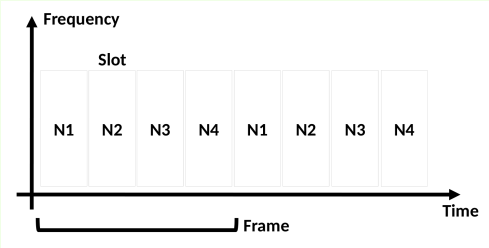
The model behind every protocol

- One channel: N stations, all transmitting and receiving on it, and no side channel on which to negotiate.
- Collisions are observable: overlapping frames are garbled and lost, and a station can tell that it happened.
- Time is continuous or slotted: slotting requires a clock all stations agree on, and not every link can provide one.
- Carrier sense, or not: whether a station can tell that the channel is busy before it starts.
- The protocols that follow differ mostly in how they answer the last two.



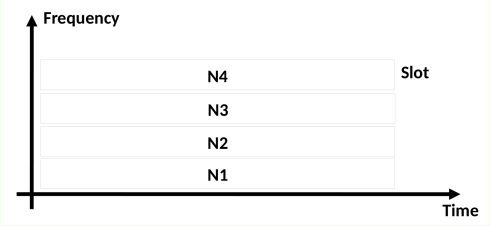
TDMA

- Time is divided into time frames, and each time frame into N slots, one assigned to each node.
- A node with a packet to send waits for its own slot and transmits during it; slots are sized to hold one whole packet.
- No collisions, and perfect fairness: everybody gets R/N during every time frame.
- Careful with the word: a TDM time frame is not a link-layer frame. Here it is a turn of the clock.



FDMA

- The R bps channel is split into N frequency bands of R/N each, one band assigned to each node.
- All the nodes transmit at the same time, permanently, each inside its own band: the channel is cut in space, not in turns.
- Same advantages as TDMA — no collisions, fair division — and exactly the same drawback.
- A node is stuck at R/N even when it is the only one with anything to send.



What a fixed split costs

- We met this argument when circuits were compared with packets: reserved capacity is wasted whenever its owner is quiet.
- Here it can be measured. On a channel of capacity C with arrival rate λ and mean frame length $1/\mu$, the mean delay is

$$T = \frac{1}{\mu C - \lambda}$$

- Split that channel into N subchannels of C/N, each receiving λ/N, and the same computation gives

$$T_N = \frac{1}{\mu(C/N) - \lambda/N} = N\,T$$

- Partitioning multiplies the mean delay by N: it is the difference between one queue for all the tills and one queue each.
- Fine where the load is steady and known. Wrong where it is bursty — and computer traffic is bursty.

CDMA

- Neither time nor frequency: each node is given a code, and uses it to encode every bit it sends.
- All the nodes transmit at once, on the whole band, all the time, and the receivers still separate them.
- Used where the medium cannot be cut cleanly: military links, for its resistance to jamming, and cellular telephony.
- Complex to implement, and it does not scale well with the number of nodes.



Codes that do not interfere

- Station A multiplies its bits by its own code c_A, station B by c_B, and the medium adds the two signals.
- The codes are chosen orthogonal: the inner product of two different codes is zero, and of a code with itself is one.
- A receiver that wants to hear B multiplies what arrives by c_B:

The cross term disappears

$$(c_A + c_B) \cdot c_B = \underbrace{c_A \cdot c_B}_{=0} + \underbrace{c_B \cdot c_B}_{=1} = 1$$

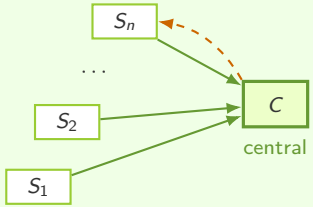
- B's bit survives and A's vanishes, although both stations transmitted simultaneously. The collision never happens.

Transmit, and gamble

- In random access nothing is assigned in advance: a node with a frame transmits it at the full rate R of the channel.
- Collisions are not prevented, they are repaired. Every node involved waits a random delay and transmits again.
- The delay must be random and drawn independently. Two stations waiting the same fixed time would collide forever.
- If the two draws differ enough, one of them sneaks its frame through; if they do not, there is another collision and another draw.
- The whole family lives on this bet. What follows is a series of ways to improve the odds.

ALOHA

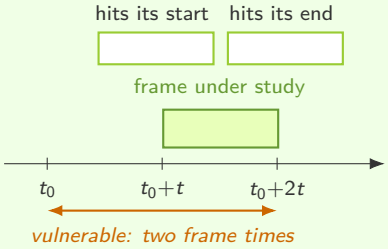
- Hawaii, 1970: terminals scattered over several islands, no cable worth laying, and one central station in Honolulu.
- The rule: transmit as soon as you have a frame. There is nothing else to the protocol.
- The central station rebroadcasts every frame it receives. A station that hears its own frame come back knows it got through.
- Silence means a collision: wait a random time, send it again. The acknowledgement is the echo, and it costs nothing extra.



C rebroadcasts what it receives: hearing your own frame come back is the acknowledgement

The vulnerable period

- Let t be the time to transmit one frame. The shaded frame survives only if nobody else starts within a window of $2t$ around it.
- Anything begun in the frame time before it collides with its beginning; anything begun during it collides with its end.
- One overlapping bit is enough: the check discards the frame either way, and a graze costs as much as a head-on hit.



At best eighteen per cent

- Call G the number of transmission attempts per frame time, new frames and retransmissions together, and assume they arrive at random.
- The vulnerable window is two frame times long, so the probability that nothing else starts in it is e^{-2G}, and the throughput is

$$S = G\,e^{-2G}$$

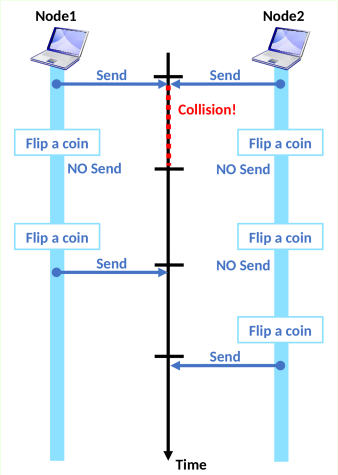
- The maximum is at $G = 0.5$, where $S = 1/2e \approx 0.18$: with everybody transmitting at will, 18% of the channel.
- Past the peak, offering more load gives less throughput — the channel collapses under its own retransmissions.
- That is the price of assuming nothing at all: no clock, no listening, no coordination of any kind.

Slotted ALOHA

- Time is divided into slots of L/R seconds — exactly one frame — and the stations are synchronised to the slot boundaries.
- A station with a fresh frame waits for the next boundary and then transmits the whole frame inside the slot.
- If two frames collide, every station involved detects it before the slot ends.
- Each colliding station then retransmits in each following slot with probability p: it tosses a biased coin, slot by slot, until the frame gets through.
- A single active node still gets the full R; the protocol is still decentralised and still trivial to implement.

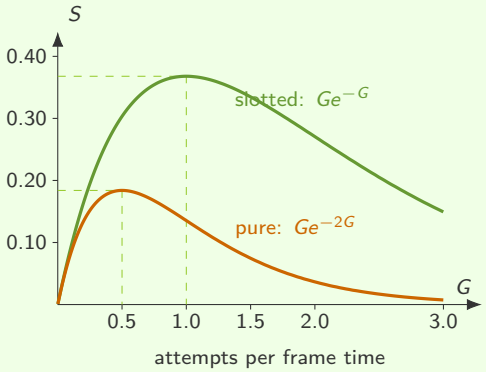
Colliding, flipping a coin, retrying

- Both nodes start at the same boundary, so the frames overlap exactly and both are lost.
- In every slot that follows, each of them tosses its coin: heads, retransmit; tails, skip this slot and toss again.
- In the picture both skip once, then node 1 sends alone and succeeds, and node 2 succeeds two slots later.
- The coins are independent, which is the only reason the two stations ever come apart.



Slots double the throughput

- Nothing can now overlap a frame partially, so the vulnerable window shrinks from two frame times to one:
$$S = G\,e^{-G}$$
- The peak moves to $G = 1$ and rises to $1/e \approx 0.37$ — exactly twice pure ALOHA.
- At the peak: 37% of slots successful, 37% empty, 26% collisions. A 100 Mb/s channel doing 37.



What slotting does not fix

Two stations decide to transmit in the middle of the same slot. What happens?

Both wait for the next boundary, both start on it, and both frames are destroyed. Slotting does not remove collisions.

- What it removes is the partial overlap: a frame can no longer be ruined by one that began a moment earlier.
- Transmissions are now either exactly aligned or entirely disjoint, and halving the vulnerable window is the whole gain.
- The price is the fifth assumption of the model: a clock every station agrees on. On a long link, or a radio one, that is not free.

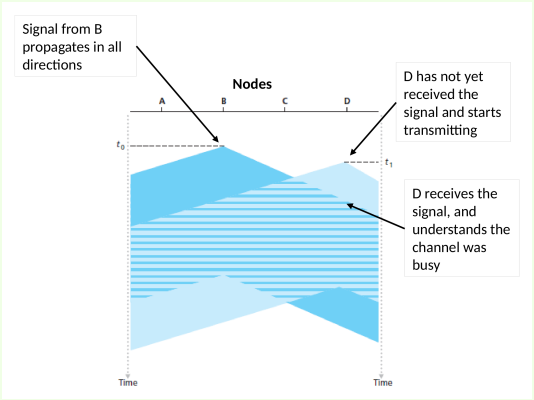
Listen before you speak

- Both ALOHA protocols transmit blind: a station decides without any regard for what is on the channel at that moment.
- Carrier sense: check the channel before transmitting. If a frame is going by, wait until it is over.
- Collision detection: keep listening while transmitting. If a collision starts, stop at once rather than finish a frame that is already lost.
- The two are the rules of polite conversation, and the two families named after them are CSMA and CSMA/CD.

If every station listens first, why would a collision ever happen?

So why do collisions still happen?

- At t_0 node B senses an idle channel and starts. Its bits travel at nearly the speed of light — but not instantly.
- At t_1 node D has a frame. B's signal has not reached it yet, so D senses idle and transmits too.
- The channel was busy and D could not know. The propagation delay is the entire reason.
- The longer the medium, the wider the window in which both stations can be wrong.



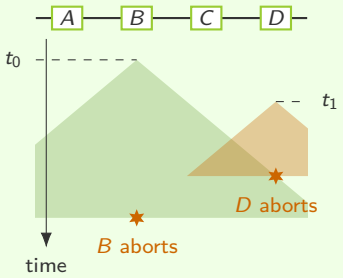
What to do when the channel is busy

Policy	On a busy channel	What it costs
1-persistent	keep listening, transmit the instant it goes idle	everyone who waited starts together: collisions
Non-persistent	wait a random time, then sense again	idle channel while everybody is waiting: delay
p-persistent	when idle, transmit with probability p	needs slots; p has to be chosen for the load

- The smaller p, the fewer stations start together: higher throughput under load, more delay when the channel is empty.
- The greedy one is the worst of the three under load — and it is the one classic Ethernet uses.

Detecting, and stopping

- Pure CSMA: sense, and transmit if idle. Nothing more: the sender never notices a collision, and the loss is discovered by whoever asks for the data again.
- CSMA/CD: the station listens to its own transmission and aborts the moment it hears interference: the channel is freed after a fraction of a frame.
- Detection is analog — compare what comes back with what goes out. Easy on cable; on radio the echo can be a millionth of the signal.



The adapter's five steps

1. Take the datagram from the network layer, build the frame, put it in the adapter's buffer.
2. Sense the channel: if there is no signal energy, start transmitting; if there is, wait until there is none.
3. While transmitting, keep watching for signal energy coming from other adapters.
4. Whole frame sent with nothing heard: done. Energy heard: abort immediately, mid-frame.
5. After aborting, wait a random amount of time and go back to step 2.

- Notice where this runs: in the adapter, in hardware. The operating system is never told that any of it happened.

Binary exponential backoff

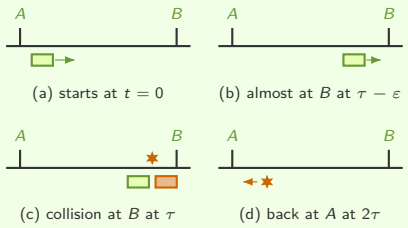
- How long should the random wait be? A short interval with many stations means they collide again; a long one with two stations leaves the channel idle for nothing.
- The interval must follow the contention, and the only measure a station has is how many collisions this frame has already had.
- After n collisions on a frame, the station draws $K \in \{0, 1, \ldots, 2^n - 1\}$ and waits K slot times. The range doubles at every failure.
- In Ethernet: a slot time is 512 bit times ($51.2\,\mu s$ at 10 Mb/s), n is capped at 10, and after 16 collisions the adapter gives up and reports failure upwards.
- That slot time is not an arbitrary number. The next two frames are where it comes from.

When can a station be sure it has the channel?

A has been transmitting for a while and has heard nothing. Is the channel hers?

Let τ be the propagation time between the two farthest stations. A cannot be sure until she has transmitted for 2τ.

- The worst case: B starts at $\tau - \varepsilon$, an instant before A's signal reaches her.
- B detects the collision almost at once, at τ — but the noise burst gets back to A only at 2τ.
- So contention is a slotted ALOHA with slots 2τ wide, except that a slot won is followed by a whole frame.

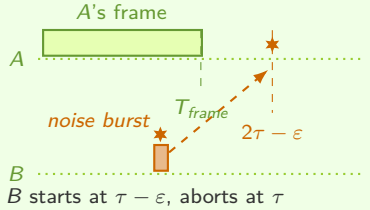


A frame that ends too soon

What if the frame takes less than 2τ to transmit?

A has finished before the noise burst arrives, and a station that is no longer transmitting is no longer listening to itself.

- A's frame is over at T_{frame}; B's noise reaches her at $2\tau - \varepsilon$, which is later.
- A concludes that the transmission succeeded. It did not: there was a collision, and the CD in CSMA/CD did nothing.
- The frame is lost in silence, and whoever finds out, higher up, pays a full timeout for it.



A has stopped transmitting, so it never hears it

Hence a minimum frame length

- The fix is to forbid short frames: if every transmission lasts more than 2τ, a station is always still transmitting when the noise gets back.
- The bound comes from the physical network. Classic 10 Mb/s Ethernet, 2500 m of cable and four repeaters: a worst-case round trip of nearly 50 μs.
- At 10 Mb/s a bit takes 100 ns, so 500 bits — rounded up for safety to 512 bits, that is 64 bytes.
- The standard therefore fixes a minimum frame of 64 bytes and adds a pad field to stretch anything shorter up to it.
- A field in a frame format that exists only because of the speed of light. We take that format apart when we get to Ethernet.

How well CSMA/CD does

- With d_{prop} the worst-case propagation time and d_{trans} the time to send a maximum-size frame:

$$\text{efficiency} \;=\; \frac{1}{1 + 5\, d_{prop}/d_{trans}}$$

- Short medium: as $d_{prop} \to 0$ colliding stations abort at once and nothing is wasted: efficiency $\to 1$.
- Long frames: as d_{trans} grows, a station that seizes the channel holds it for a long time and the contention is amortised.
- What hurts is the ratio: a long cable carrying short frames. It is the same ratio that fixed the minimum frame length.
- And today: none of this runs on a switched full-duplex link, where a station has the wire to itself. CSMA/CD is the protocol of shared, half-duplex Ethernet — which is where switches come in.

The property nobody had

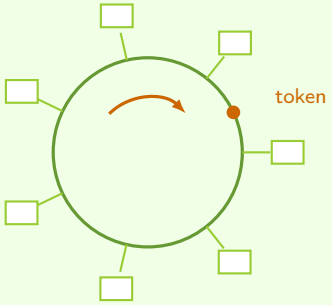
- Two of the four desiderata: full R for a lone station, and R/M each when M are active.
- Partitioning: gives the second and fails the first — the lone station is stuck at R/N while the other bands sit empty.
- Random access: gives the first and fails the second — under load the throughput does not settle at R/M, it collapses.
- The third family arbitrates before anyone transmits: a station may not send unless it is its turn, so there are no collisions at all.
- The price is always the same. Turns have to be organised, and the organisation is one more thing that can break.

Polling

- One node is the master. It invites each of the others in turn to transmit, up to a maximum number of frames each.
- No collisions and no empty slots: nothing is transmitted that nobody asked for, so the efficiency is high.
- Polling delay: even a single active station pays for the master's round through all the silent ones.
- Centralised: if the master fails, the whole channel stops. That is the desideratum this family gives up.
- Bluetooth works this way: the device that opened the connection polls the others.

Token passing

- No master. A small special frame — the token — circulates among the nodes in a fixed order.
- Holding it is permission to transmit: a node with nothing to send forwards it at once, one with frames sends and then passes it on.
- Decentralised, efficient, and with a bounded worst-case wait — which no contention protocol can promise. The ring may be only logical: physically a bus, logically an order.
- The token is the weak point: lost, or held by a station that has failed, it stops everything until it is recovered.



All three, in one network

Cable access

Several thousand cable modems share the coaxial network of a neighbourhood, with one CMTS at the head end. DOCSIS is the link-layer protocol that governs it.

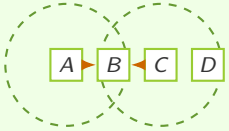
- Partitioning: FDM cuts the cable into downstream and upstream channels, each of them a broadcast channel of its own.
- No problem downstream: only the CMTS transmits there, so there is nothing to arbitrate.
- Taking turns upstream: time is cut into mini-slots, and the CMTS grants them to named modems with a control message.
- Random access to ask: a modem requests a mini-slot in slots reserved for requests. It cannot sense the channel, so it infers a collision from the grant that never comes — and backs off exponentially.

Where carrier sense fails

On radio, can a station hear everything that could collide with its frame?

No. What matters is whether the signals collide at the receiver — which is exactly what the sender cannot observe.

- Hidden terminals: A and C both reach B but not each other. Both sense idle, both transmit, and the collision at B is undetectable by either.
- Exposed terminals: B is sending to A and C wants to send to D. C hears B and defers, although nobody would have been disturbed.

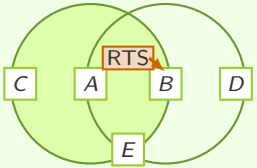


(a) A and C do not hear each other, and collide at B.

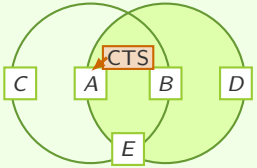
(b) C hears B and defers, although the two transmissions go opposite ways

Asking before sending

- MACA replaces listening with a reservation. A sends a short RTS to B; B answers with a CTS, and both carry the length of what is coming.
- The CTS comes from the receiver, so it reaches exactly the stations that could ruin the reception — including the ones hidden from A — and silences them.
- A station that hears only the RTS knows it is far from the receiver and may transmit elsewhere: the exposed terminal is no longer blocked for nothing. It is the idea behind the RTS/CTS exchange of 802.11.



A sends RTS to B;
C and E hear it and keep quiet



B replies with CTS;
D and E hear it and keep quiet

Summary

- A shared medium has no arbiter: the rules have to be agreed on the same channel that is being contended for.
- Partitioning: no collisions, and the share of whoever is quiet is wasted — N times the delay when the traffic comes in bursts.
- Random access: a bet, and the odds improve as the assumptions grow: 18% blind, 37% with a clock, far more with carrier sense.
- Collisions survive carrier sense because of the propagation delay; detecting them shortens the damage, and the 2τ argument fixes the minimum frame length.
- Taking turns: no collisions and a bounded wait, paid for with a master that can fail or a token that can be lost.
- All of this treats stations as anonymous. Next: how they are named, and what a real local network does with those names.